

Analysis

Algorithm	Best Case	Average Case	Worst Case
Linear Search	$O(1)$	$O(n)$	$O(n)$
Binary Search	$O(1)$	$O(\log n)$	$O(\log n)$

Linear Search is simple but takes $O(n)$ time in the average and worst cases. Binary Search requires the products to be sorted but performs much faster with $O(\log n)$ complexity. Since an e-commerce platform usually handles a large number of products, Binary Search is generally the preferred approach for faster product retrieval.